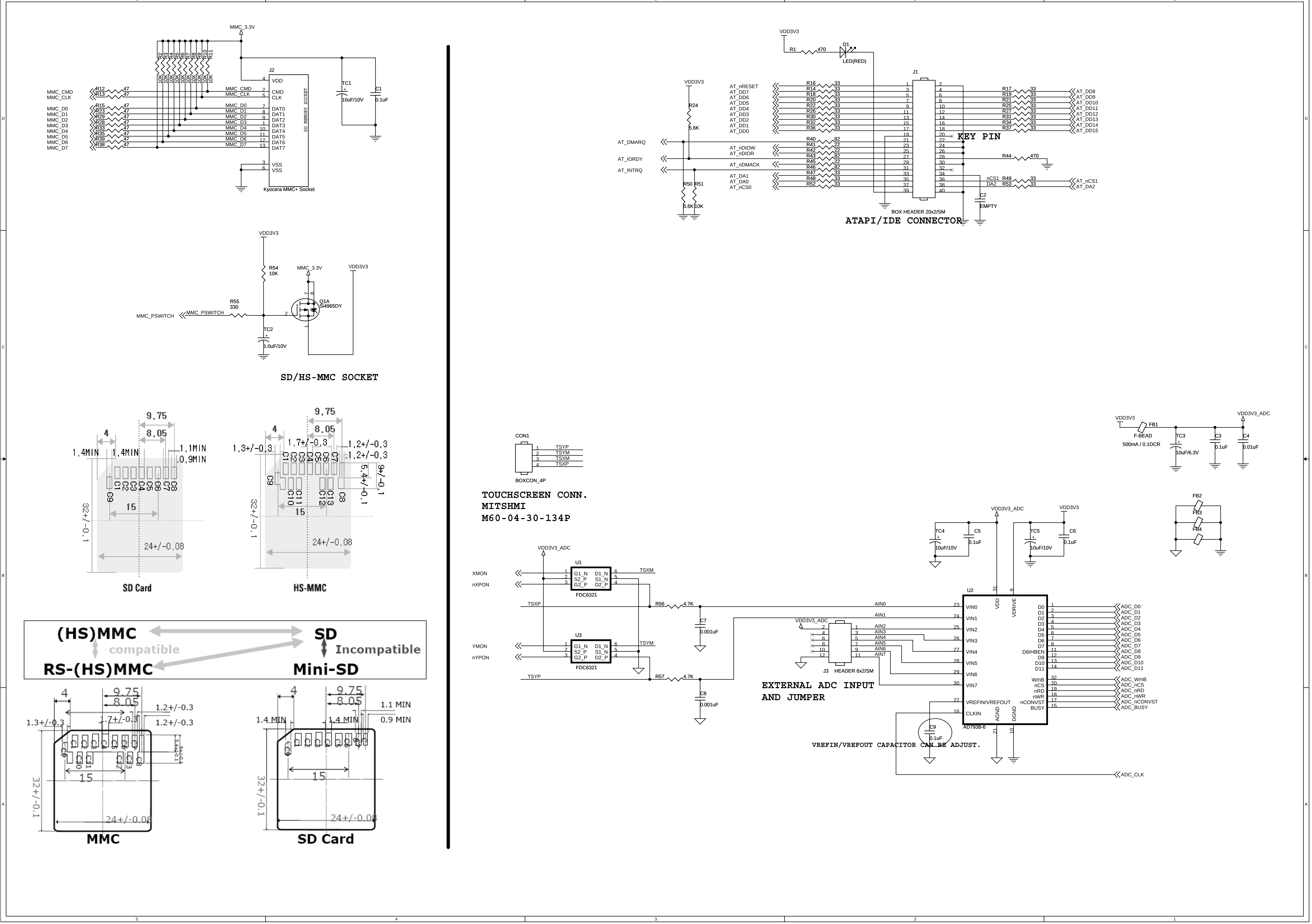

SD/HS-MMC SOCKET
MMC_3.3V
J2
Kyocera MMC+ Socket
SD MEMORY SOCKET
MMC_CMD
MMC_CLK
MMC_D0
MMC_D1
MMC_D2
MMC_D3
MMC_D4
MMC_D5
MMC_D6
MMC_D7
TC1 10uF/10V
C1 0.1uF
VDD3V3
R54 10K
R55 330
MMC_PSWITCH
Q1A Si4965DY
TC2 1.0uF/10V
ATAPI/IDE CONNECTOR
KEY PIN
BOX HEADER 20x2/SM
J1
D1 LED(RED)
R1 470
AT_nRESET
AT_DMARQ
AT_IORDY
AT_INTRQ
AT_nDIOW
AT_nDIOR
AT_nDMACK
AT_DA1
AT_DA0
AT_nCS0
AT_nCS1
AT_DA2
C2 EMPTY
TOUCHSCREEN CONN.
MITSHMI
M60-04-30-134P
CON1
BOXCON_4P
TSYP
TSYM
TSXM
TSXP
U1 FDC6321
U3 FDC6321
XMON
nXPON
YMON
nYPON
VDD3V3_ADC
R56 4.7K
R57 4.7K
C7 0.001uF
C8 0.001uF
EXTERNAL ADC INPUT
AND JUMPER
J3 HEADER 6x2/SM
U2 AD7938-6
VREFIN/VREFOUT CAPACITOR CAN BE ADJUST.
C9 0.1uF
ADC_CLK
FB1 F-BEAD 500mA / 0.1DCR
TC3 10uF/6.3V
C3 0.1uF
C4 0.01uF
FB2
FB3
FB4
TC4 10uF/10V
C5 0.1uF
TC5 10uF/10V
C6 0.1uF
9.75
8.05
4
1.4MIN
1.1MIN
0.9MIN
15
32+/-0.1
24+/-0.08
SD Card
1.3+/-0.3
1.7+/-0.3
1.2+/-0.3
5.4+/-0.1
9+/-0.1
HS-MMC
(HS)MMC
SD
compatible
Incompatible
RS-(HS)MMC
Mini-SD
MMC
SD Card

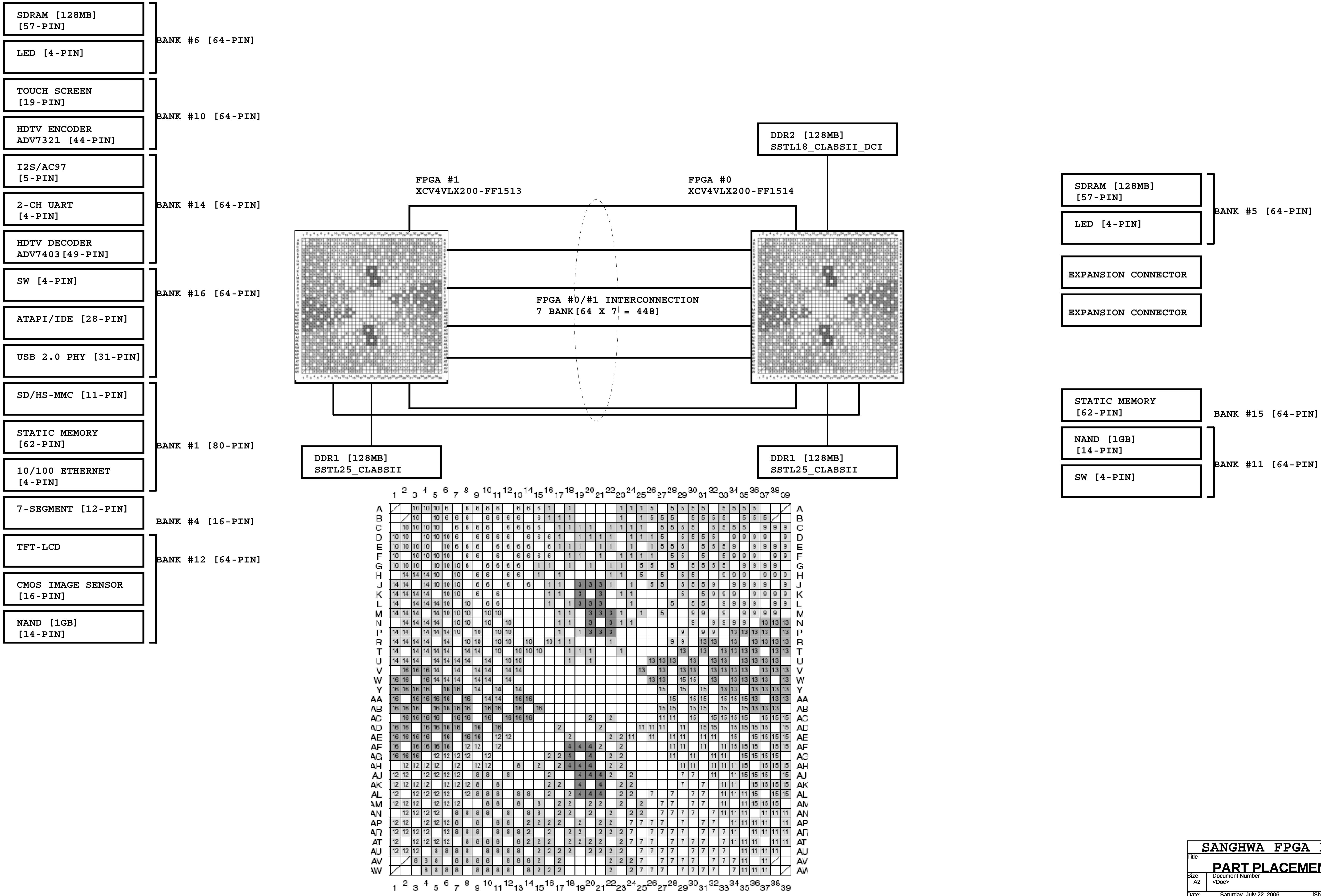
SANGHWA FPGA PLATFORM

PAGE #	DESCRIPTION
P00	TITLE PAGE (THIS)
P01	SD/MMC/TOUCH
P02	SYSTEM ACE (FPGA CONFIGURATION)
P03	NONE
P04	VIDEO
P05	MEMORY BLOCK
P06	SW/AC97
P07	ETH/USB/UART
P08	CLOCK/PLL/MULTIPLEXOR
P09	SYSTEM POWER
P10	FPGA TEST MICTOR
P11	F0 CONFIG/CLOCK
P12	F0 BANK 1/2/5/6
P13	F0 BANK 7/8/9/10
P14	F0 BANK 11/12/13/14
P15	F0 BANK 15/16/GND
P16	F0 POWER VCCO
P17	F1 CONFIG/CLOCK
P18	F1 BANK 1/2/5/7
P19	F1 BANK 7/8/9/11
P20	F1 BANK 11/12/13/15
P21	F1 BANK 15/16/GND
P22	F1 POWER VCCO

2006-7-20
DESIGNED BY DDSMURF

SANGHWA FPGA PLATFORM

Title	SANGHWA FPGA PLATFORM	
Size A3	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 1 of 26	



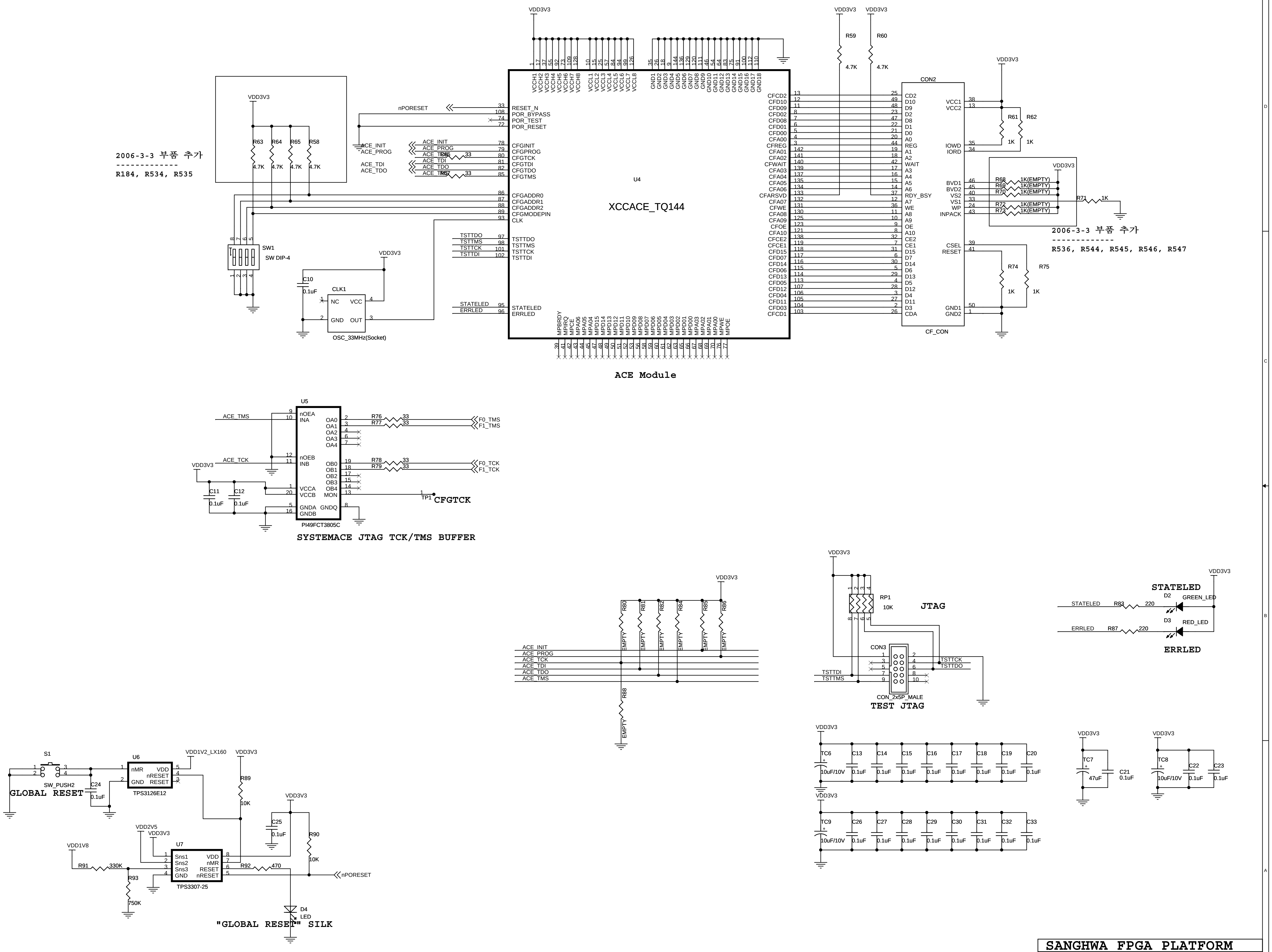
SANGHWA FPGA PLATFORM

Title

PART PLACEMENT

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 2 of 26

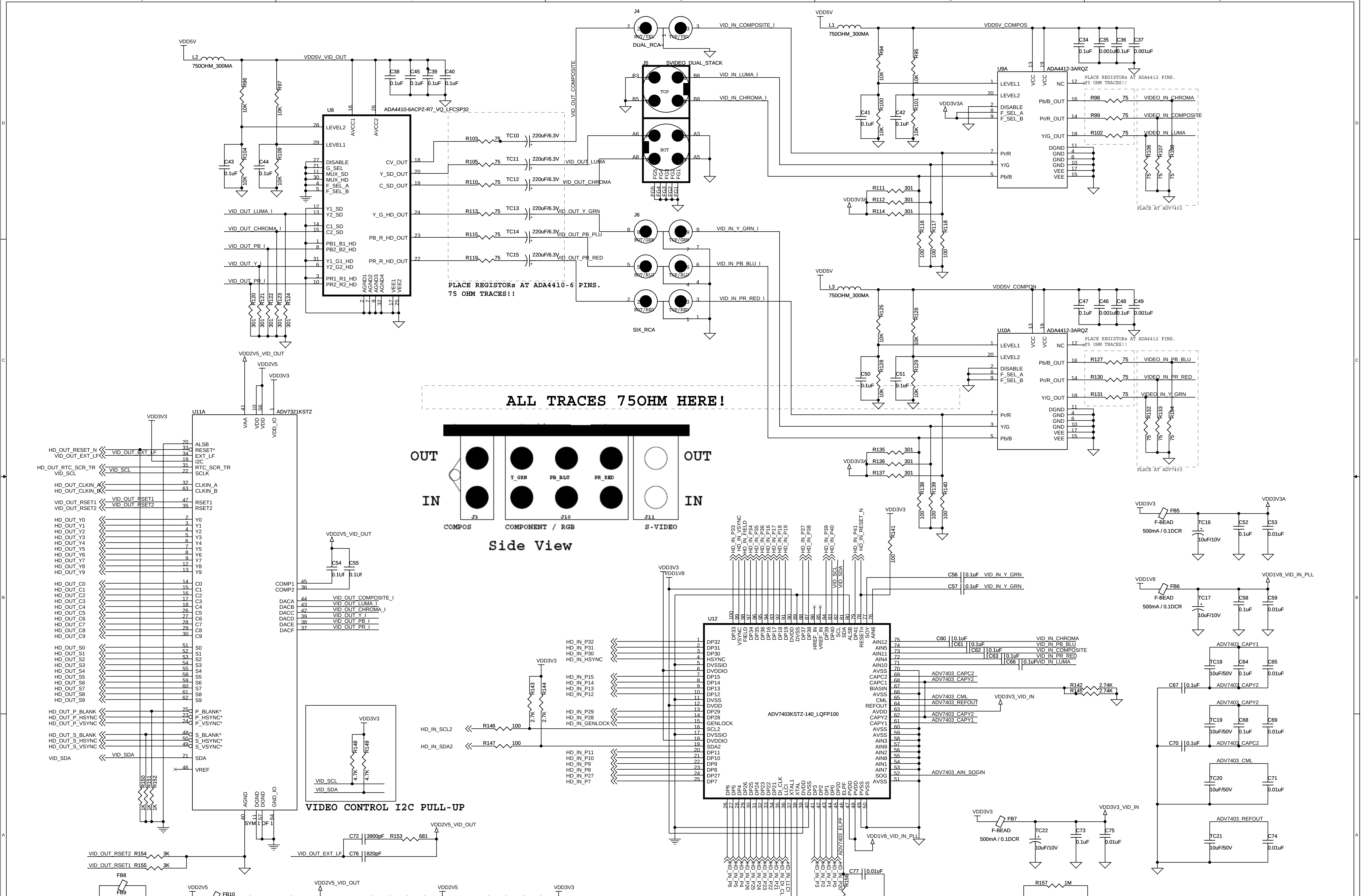


SANGHWA FPGA PLATFORM

Title: SYSTEM ACE

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 — Sheet 4 of 26

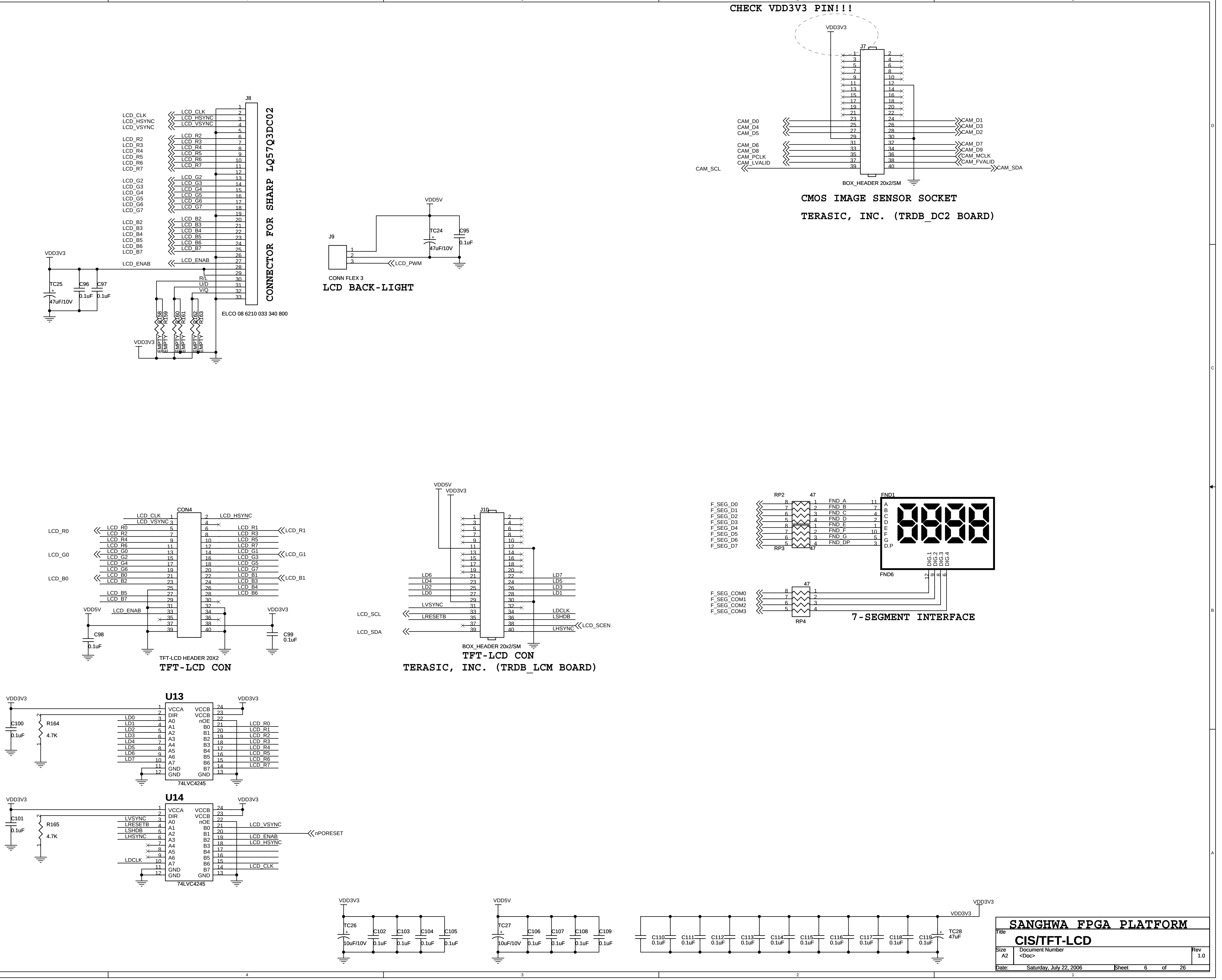


SANGHWA FPGA PLATFORM

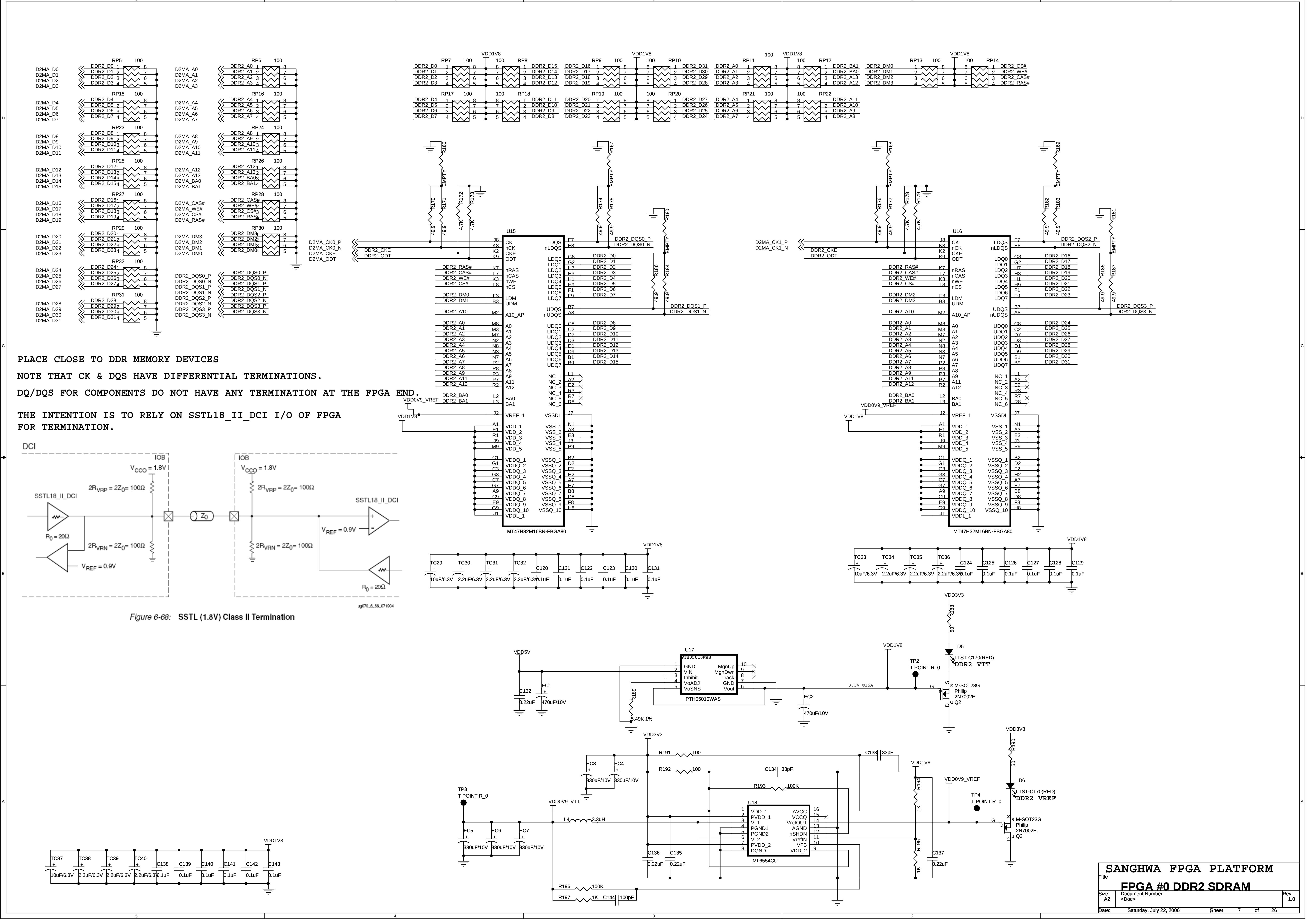
Title: HD VIDEO ENCODER/DECODER

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 5 of 26



SANGHWA FPGA PLATFORM		
Title: CIS/TFT-LCD		
Size: A2	Document Number: <Doc>	Rev: 1.0
Date: Saturday, July 22, 2006	Sheet 6 of 26	



PLACE CLOSE TO DDR MEMORY DEVICES

NOTE THAT CK & DQS HAVE DIFFERENTIAL TERMINATIONS.

DQ/DQS FOR COMPONENTS DO NOT HAVE ANY TERMINATION AT THE FPGA END.

THE INTENTION IS TO RELY ON SSTL18_II_DCI I/O OF FPGA
FOR TERMINATION.

Figure 6-68: **SSTL (1.8V) Class II Termination**

Title	SANGHWA FPGA PLATFORM
	FPGA #0 DDR2 SDRAM
Size	A2
Document Number	<Doc>
Rev	1.0
Date:	Saturday, July 22, 2006
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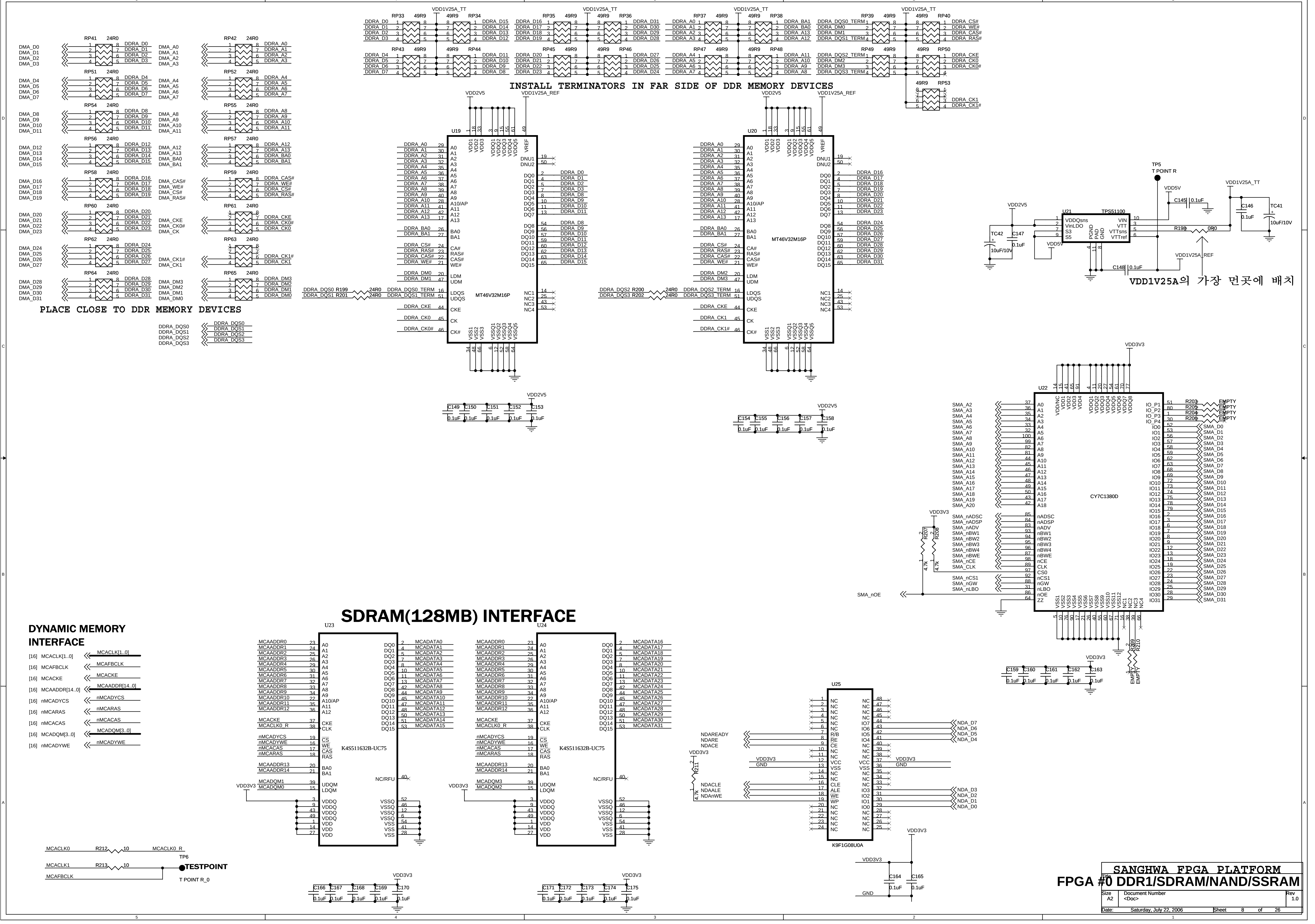
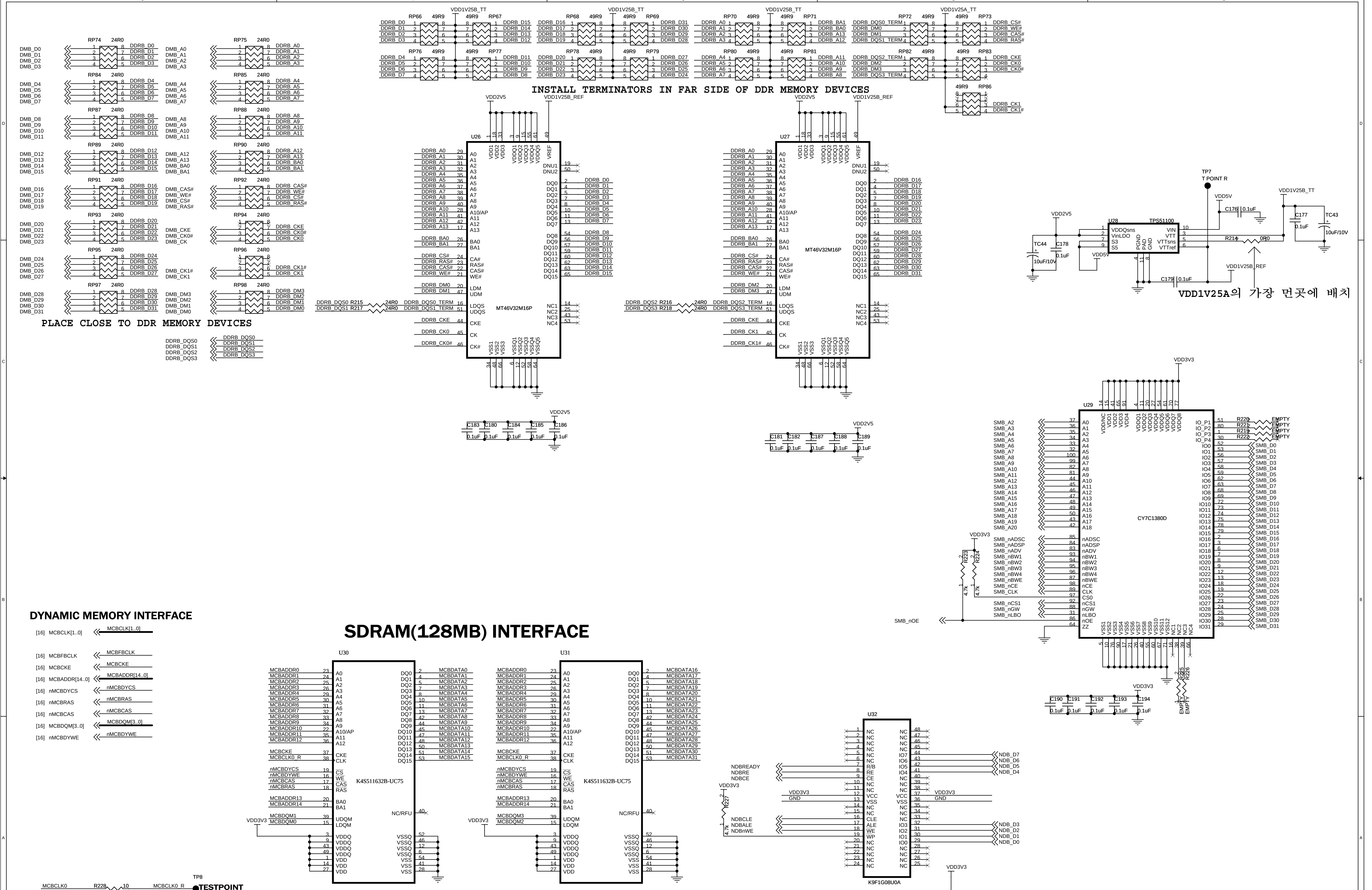


INSTALL TERMINATORS IN FAR SIDE OF DDR MEMORY DEVICES
PLACE CLOSE TO DDR MEMORY DEVICES
VDD1V25A의 가장 먼곳에 배치
SDRAM(128MB) INTERFACE
DYNAMIC MEMORY INTERFACE
MT46V32M16P
TPS51100
CY7C1380D
K4S511632B-UC75
K9F1G08U0A
TESTPOINT
SANGHWA FPGA PLATFORM
FPGA #0 DDR1/SDRAM/NAND/SSRAM
Size A2
Document Number <Doc>
Rev 1.0
Date: Saturday, July 22, 2006
Sheet 8 of 26

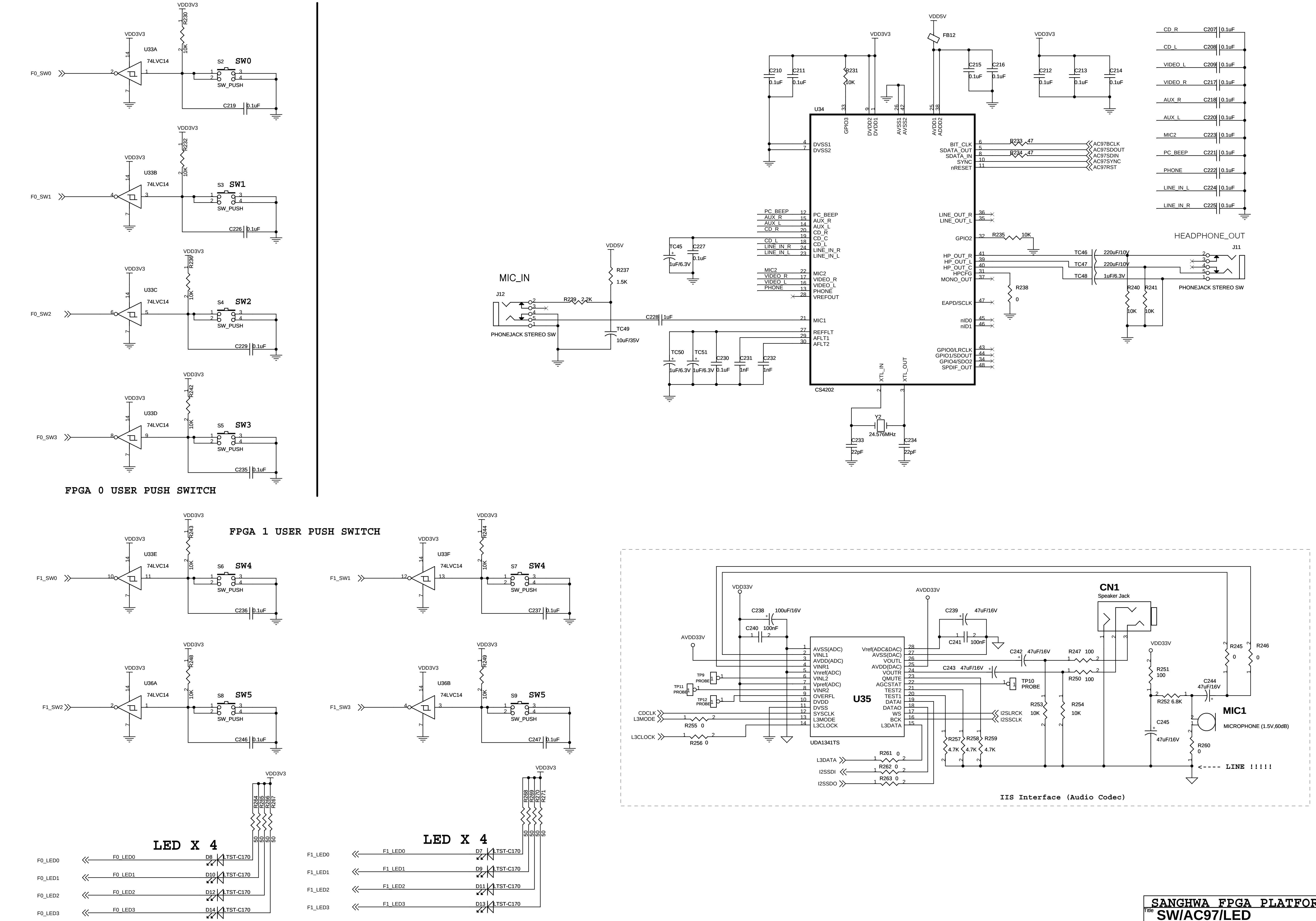


SANGHWA FPGA PLATFORM

Title: FPGA #1 DDR1/SDRAM/NAND/SSRAM

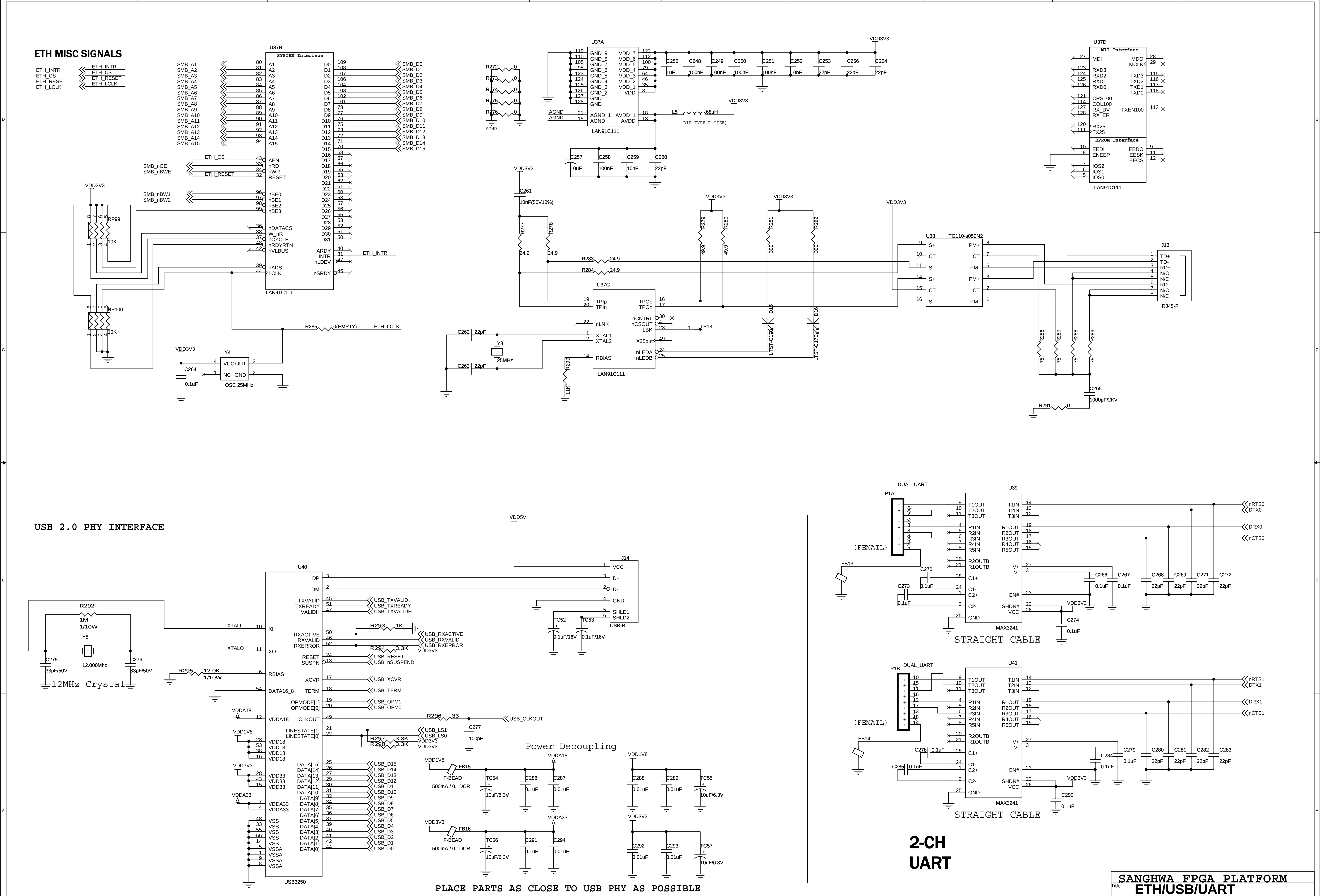
Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 9 of 26



SANGHWA FPGA PLATFORM

Title	SW/AC97/LED		
Size A2	Document Number <Doc>		Rev 1.0
Date:	Saturday, July 22, 2006	Sheet 10 of 26	

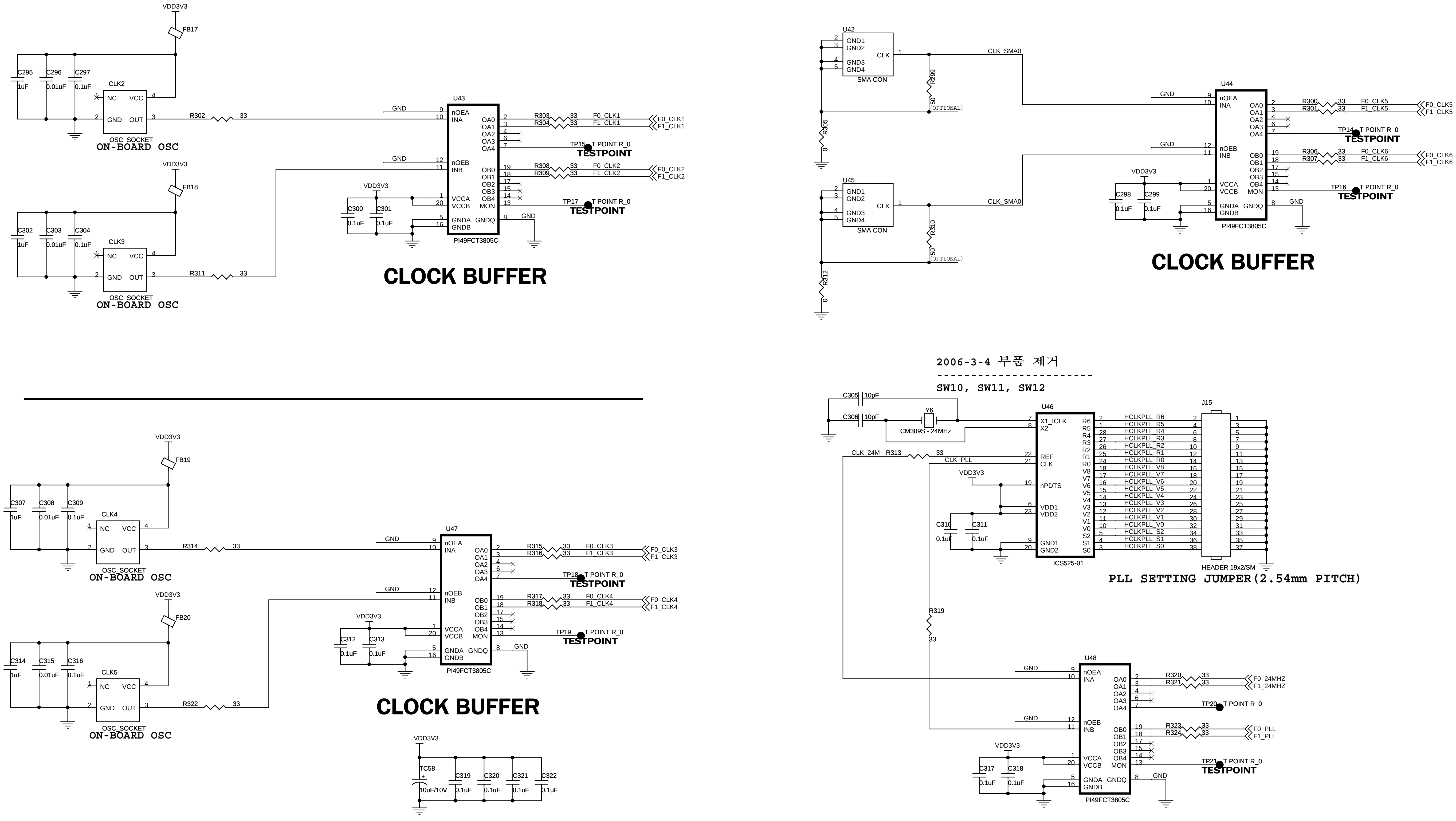


SANGHWA FPGA PLATFORM

Title: ETH/USB/UART

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 11 of 26

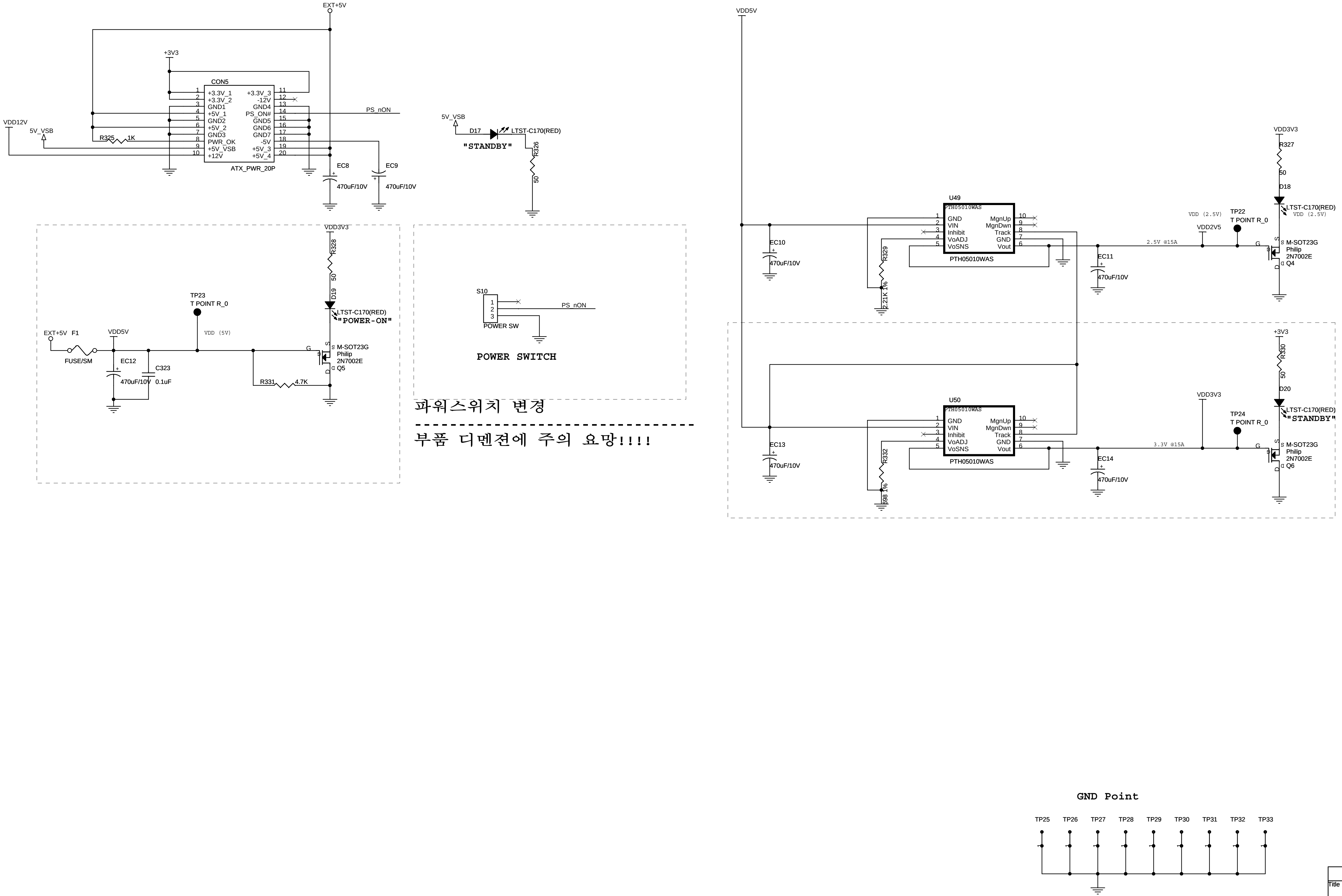


SANGHWA FPGA PLATFORM

Title CLOCK/PLL/MULTIPLEXOR

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 12 of 26



Title	SANGHWA FPGA PLATFORM SYSTEM POWER	
Size A2	Document Number <Doc>	Rev 1.0
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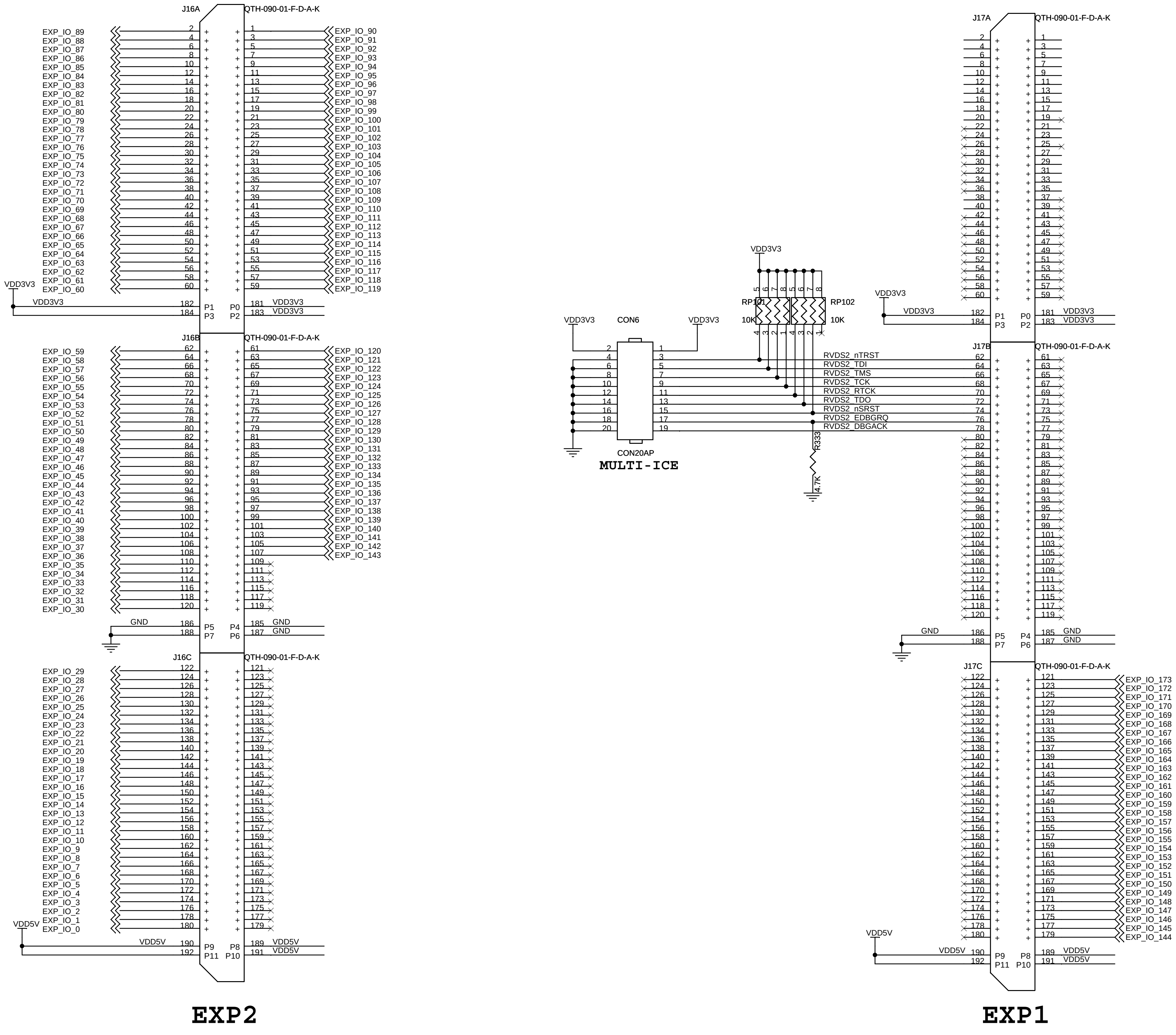
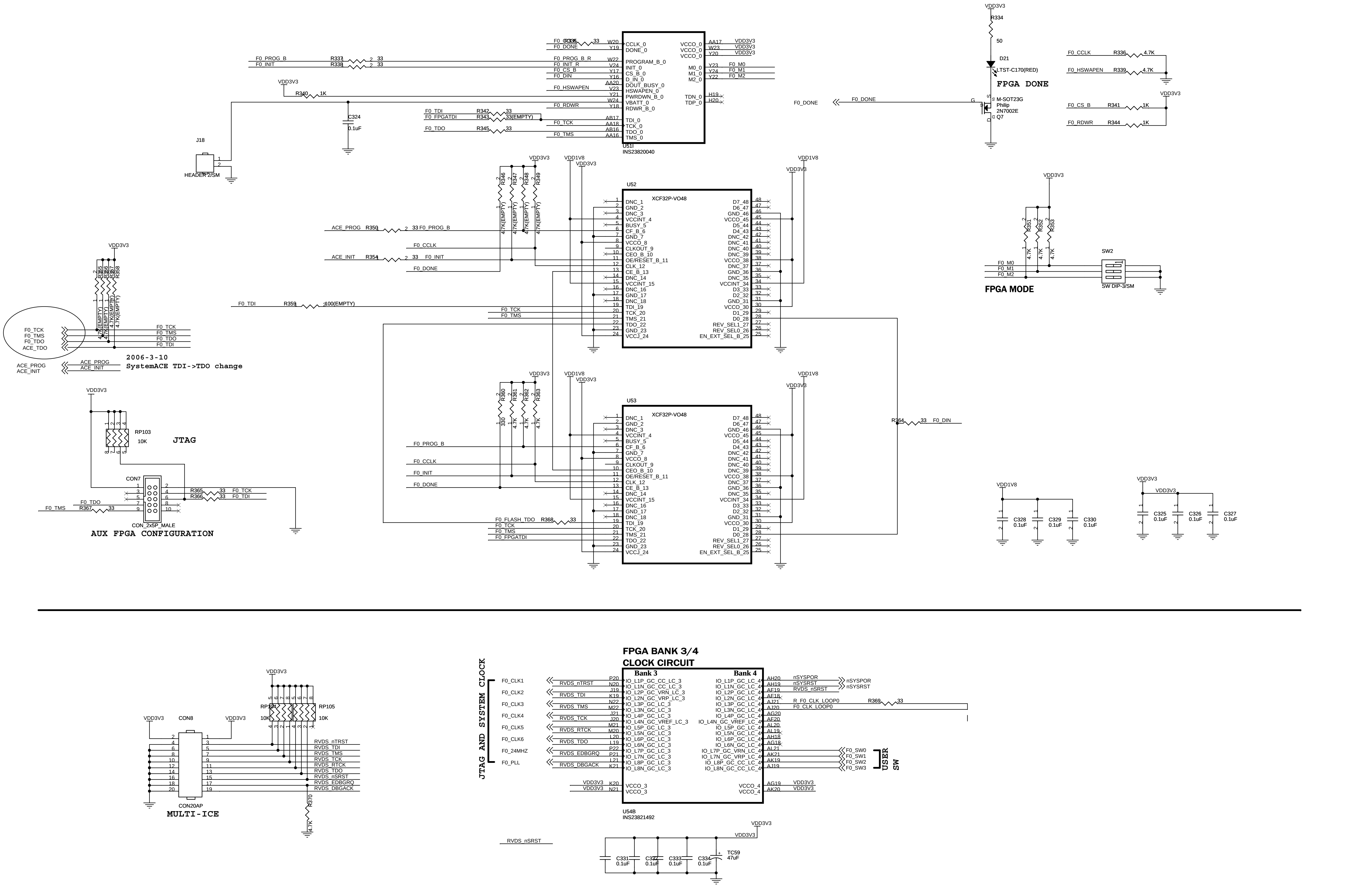
EXPANSION SLOT FOR FPGA #0



SANGHWA FPGA PLATFORM		
Title FPGA #0 EXPANSION CONNECTOR		
Size A2	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 14 of 26	

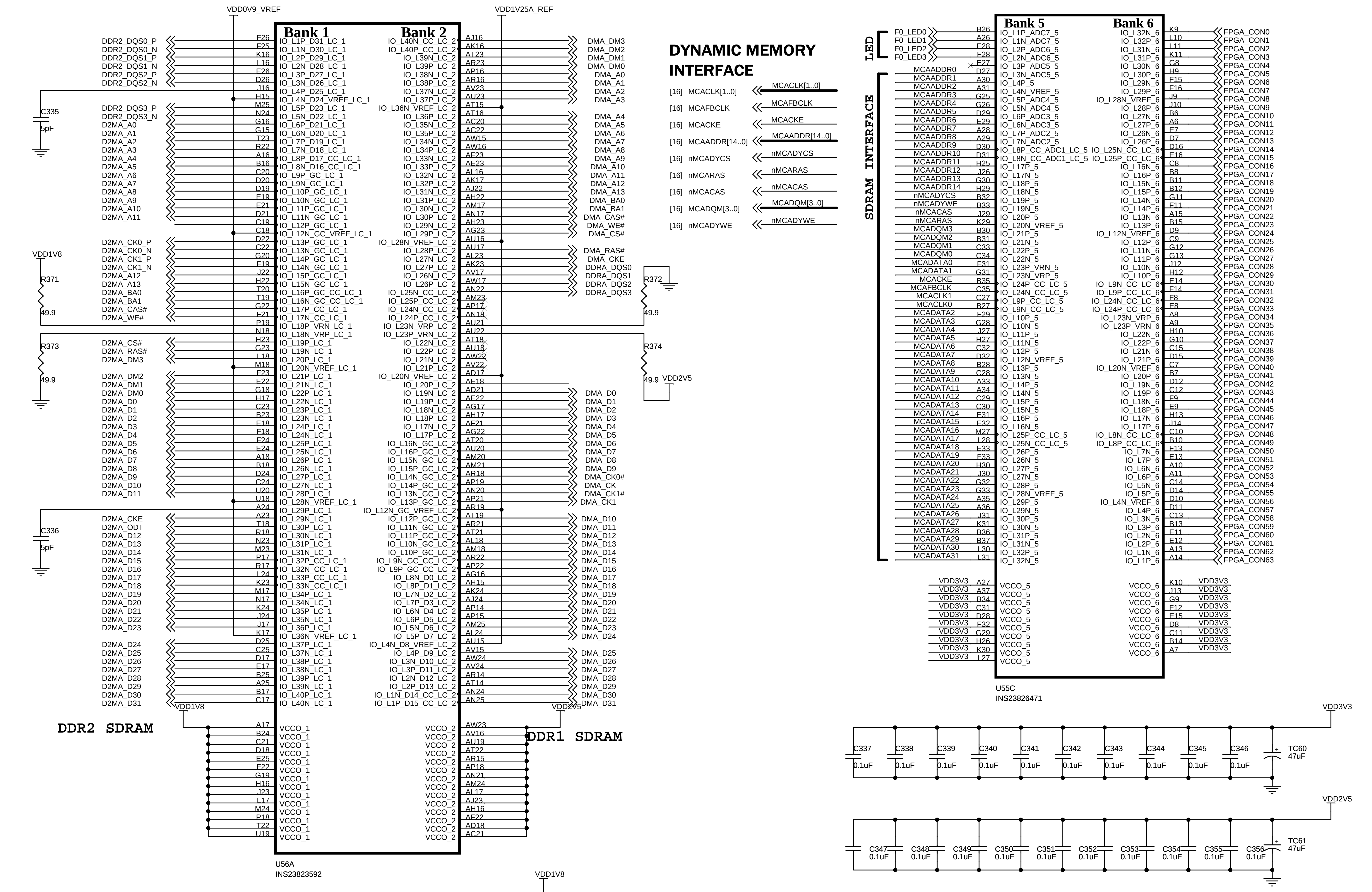


SANGHWA FPGA PLATFORM

Title **FPGA0 CONFIG/CLOCK**

Size	Document Number	Rev
A2	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 15 of 26



Title	SANGHWA FPGA PLATFORM FPGA0 BANK 1/2/5/6
Size A3	Document Number <Doc>
Rev	1.0
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FPGA INTERCONNECTION

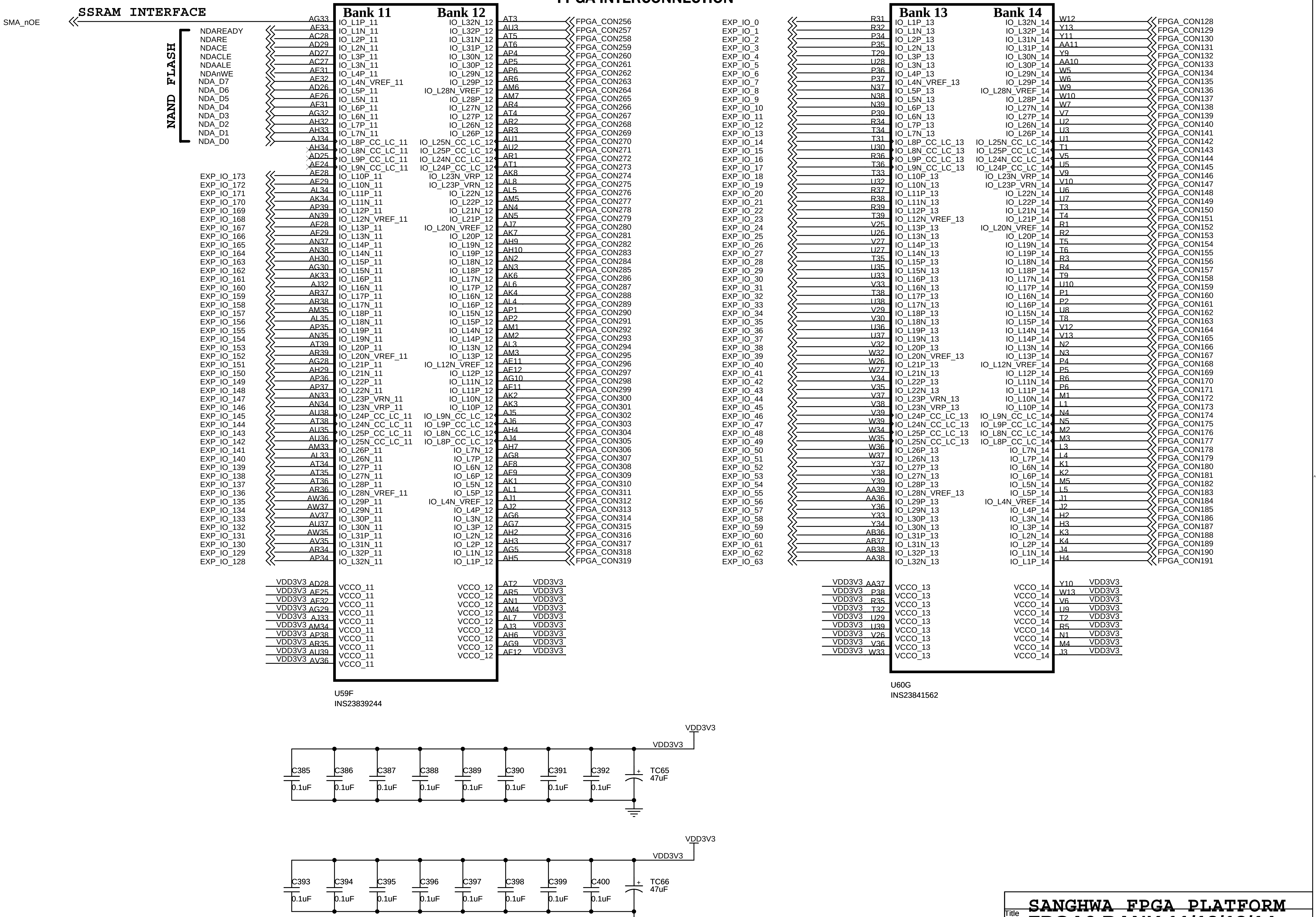
SANGHWA FPGA PLATFORM

Title: FPGA0 BANK 7/8/9/10

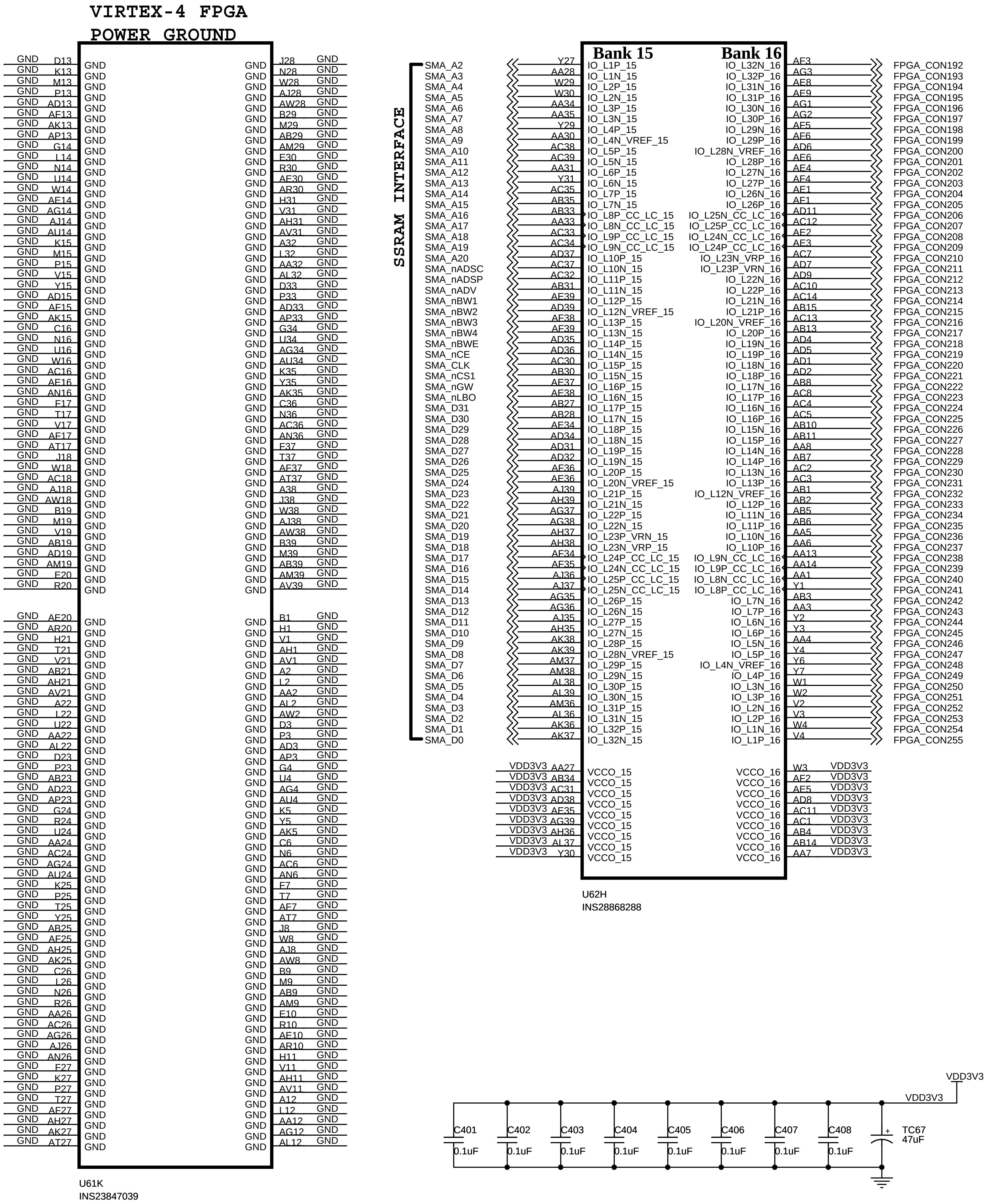
Size	Document Number	Rev
A3	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 17 of 26

FPGA INTERCONNECTION



Title	SANGHWA FPGA PLATFORM FPGA0 BANK 11/12/13/14	
Size A3	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 18 of 26	

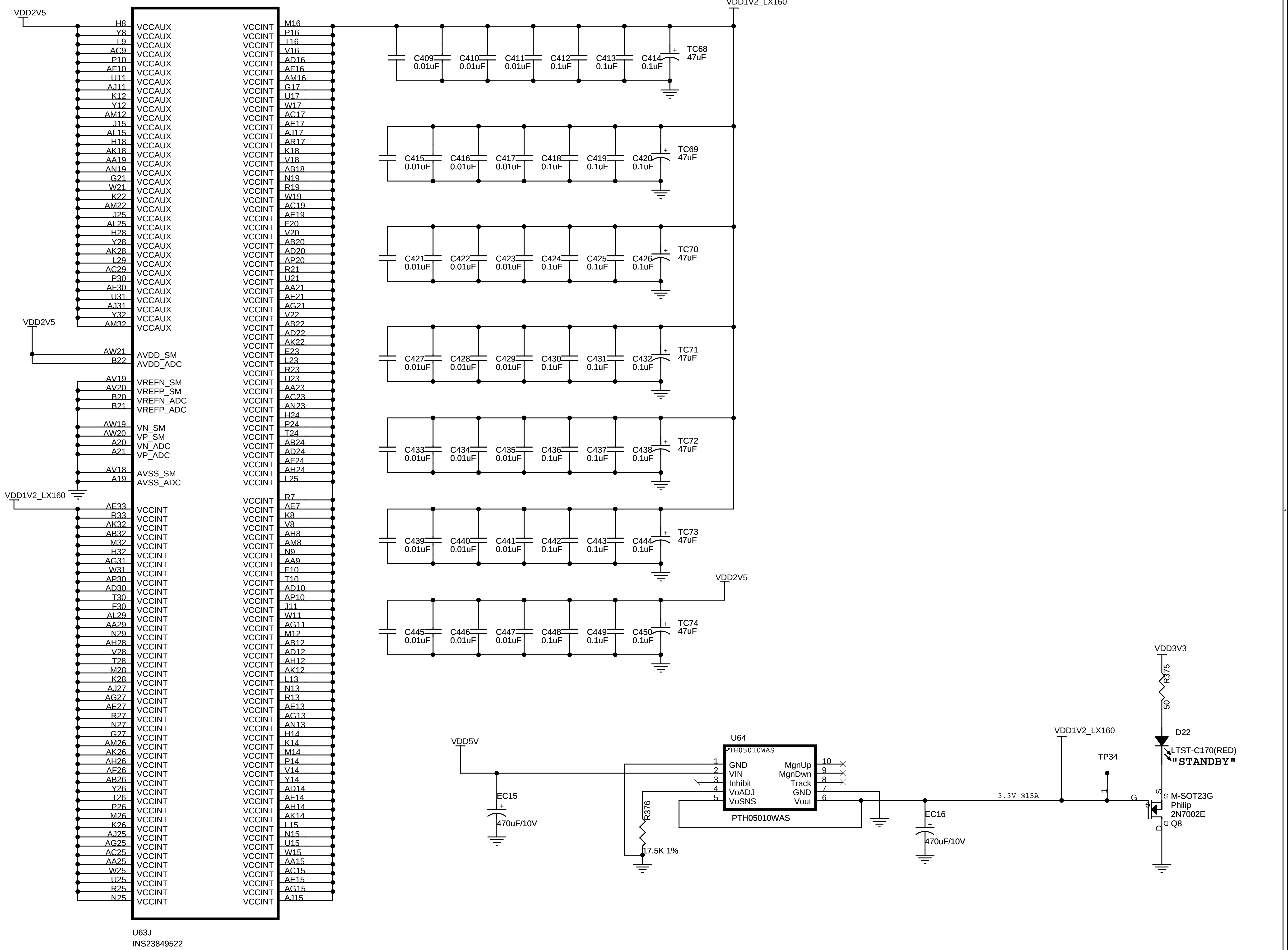


SANGHWA FPGA PLATFORM

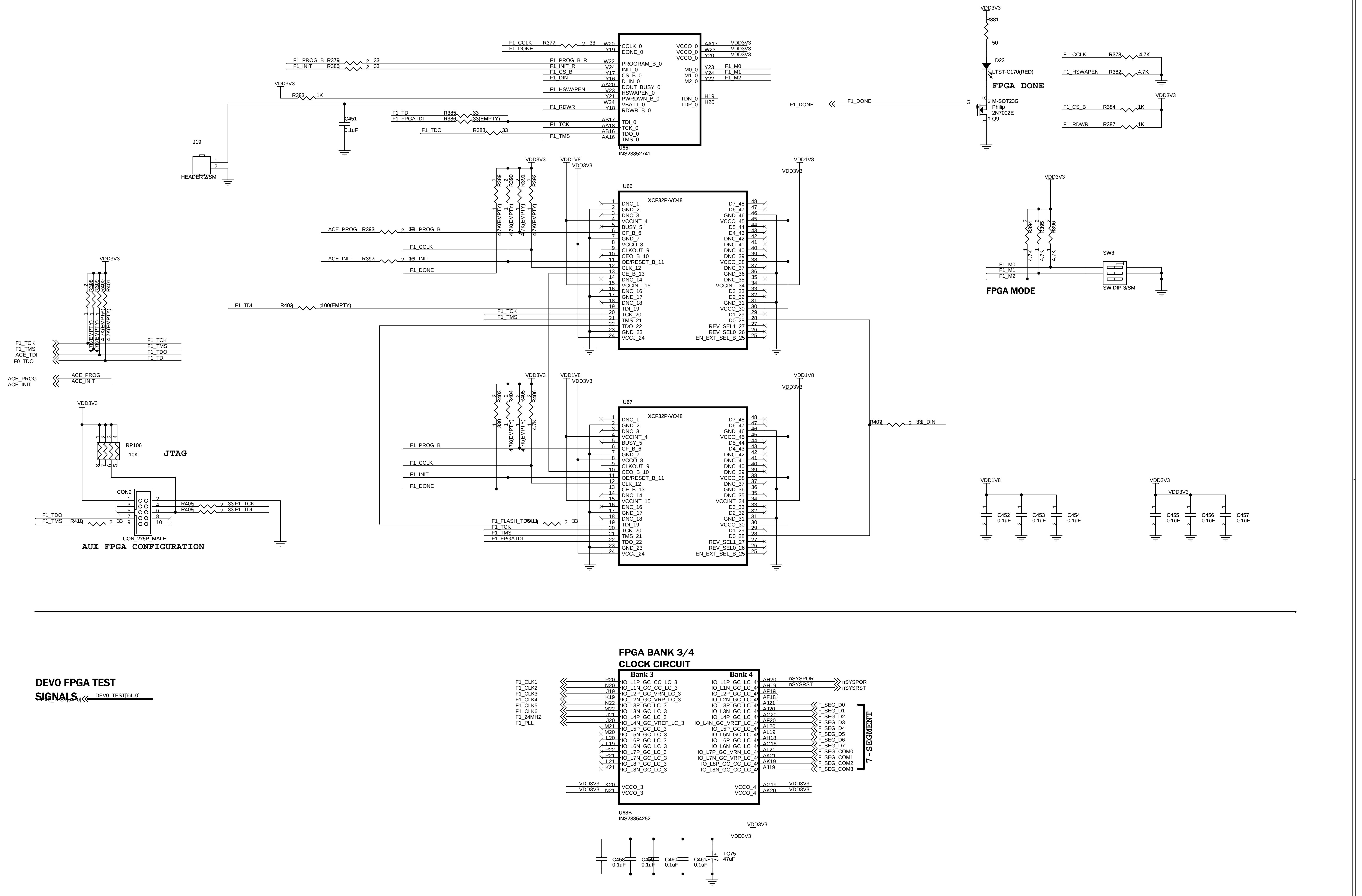
Title **FPGA0 BANK 15/16/GND**

Size	Document Number	Rev
A3	<Doc>	1.0

Date: Saturday, July 22, 2006 Sheet 19 of 26

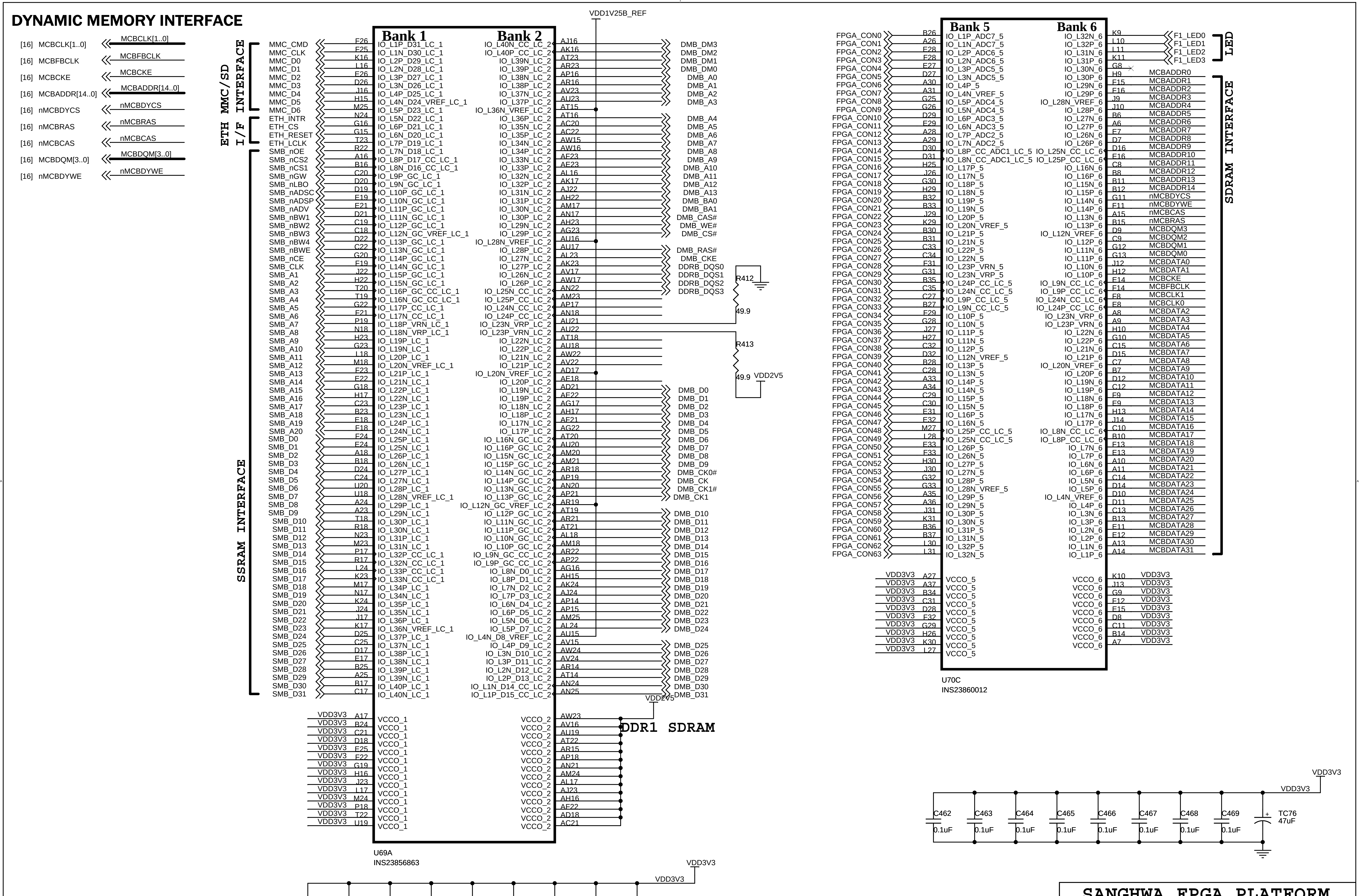


	SANGHWA FPGA PLATFORM		
Title	FPGA0 POWER VCCO		
Size A3	Document Number <Doc>		Rev 1.0
Date:	Saturday, July 22, 2006	Sheet 20 of 26	



SANGHWA FPGA PLATFORM

Title	FPGA1 CONFIG/CLOCK	
Size A2	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 21 of 26	

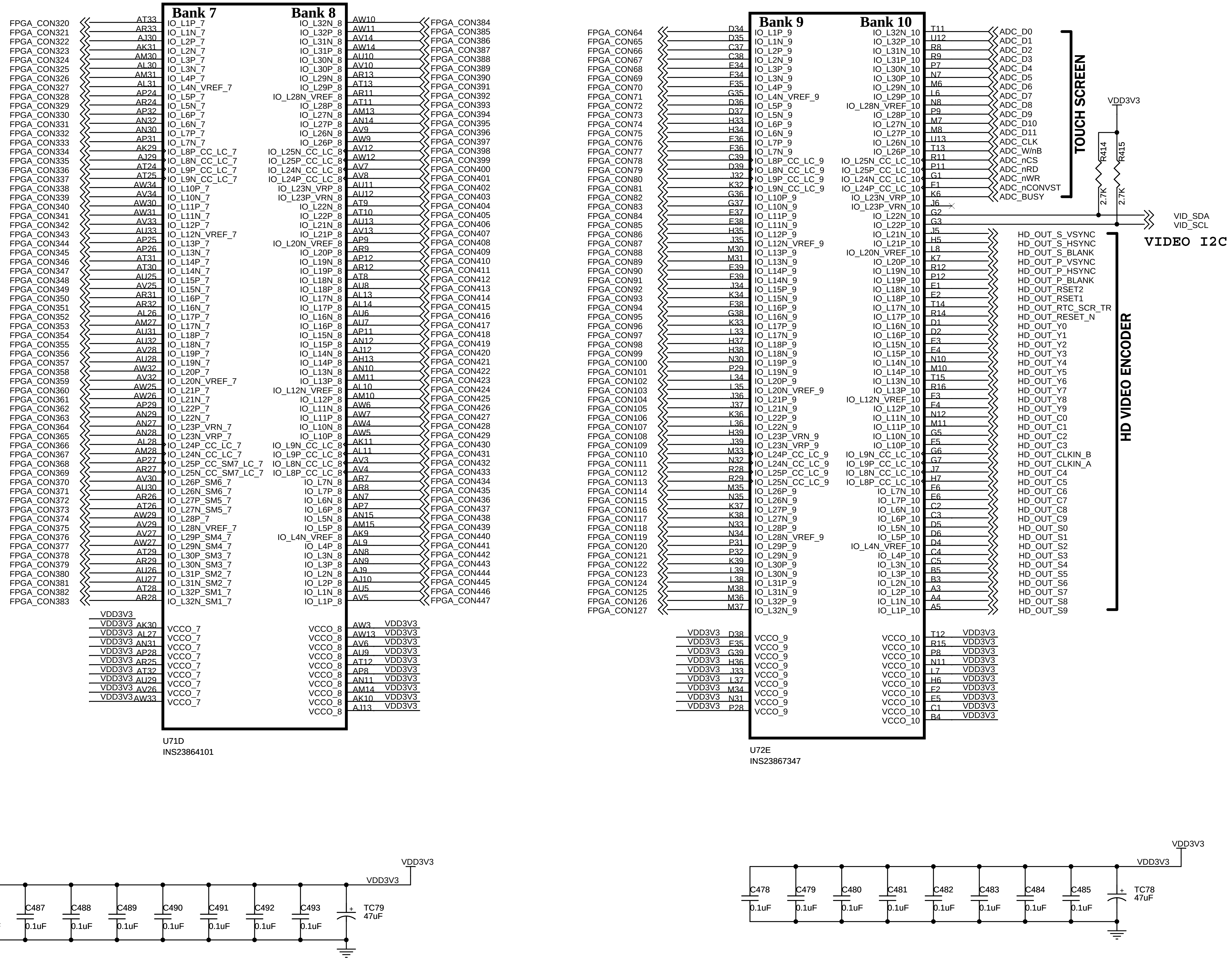


SANGHWA FPGA PLATFORM

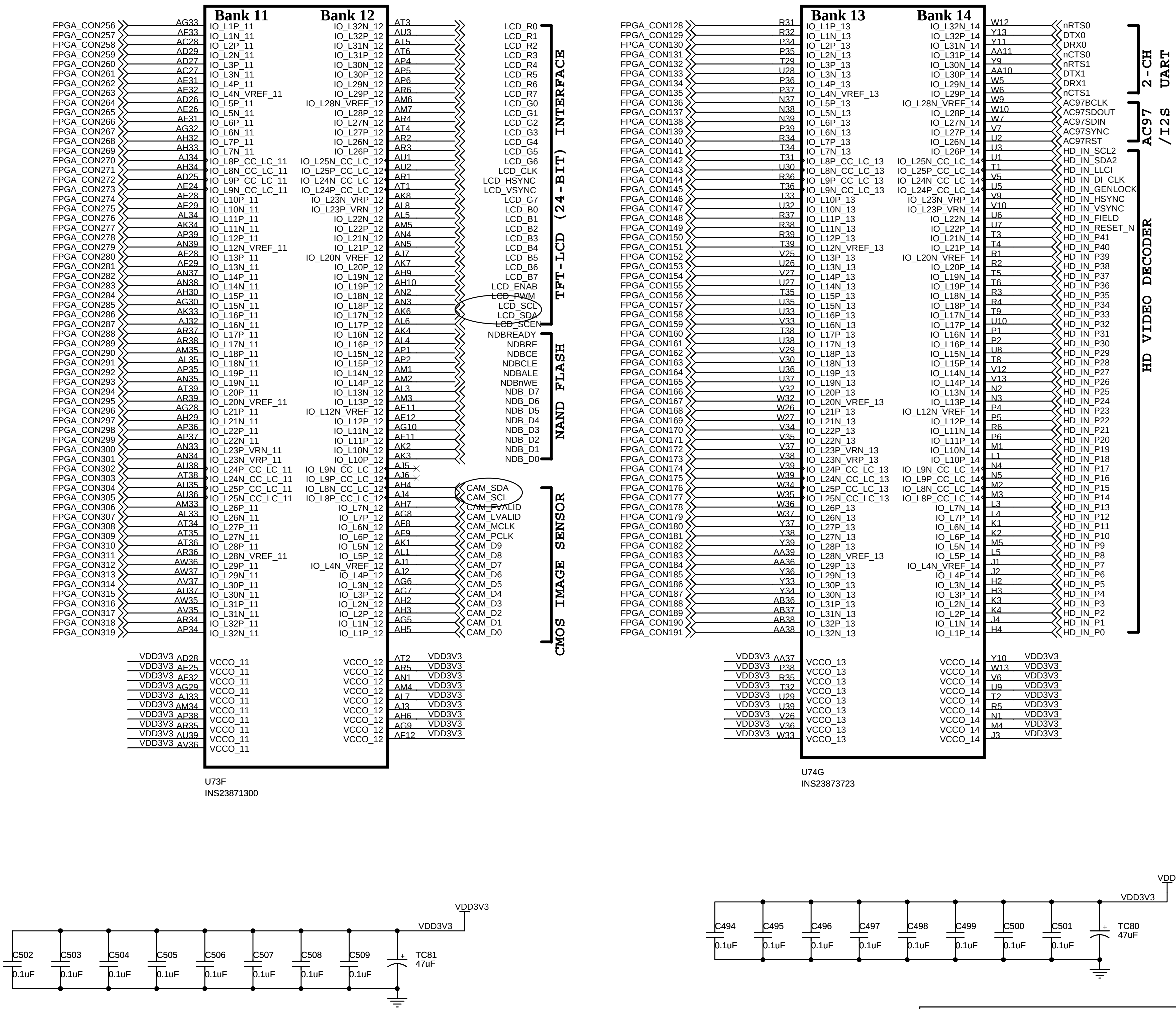
Title: FPGA1 BANK 1/2/5/6

Size	Document Number	Rev
A3	<Doc>	1.0

Date: Saturday, July 22, 2006 | Sheet 22 of 26

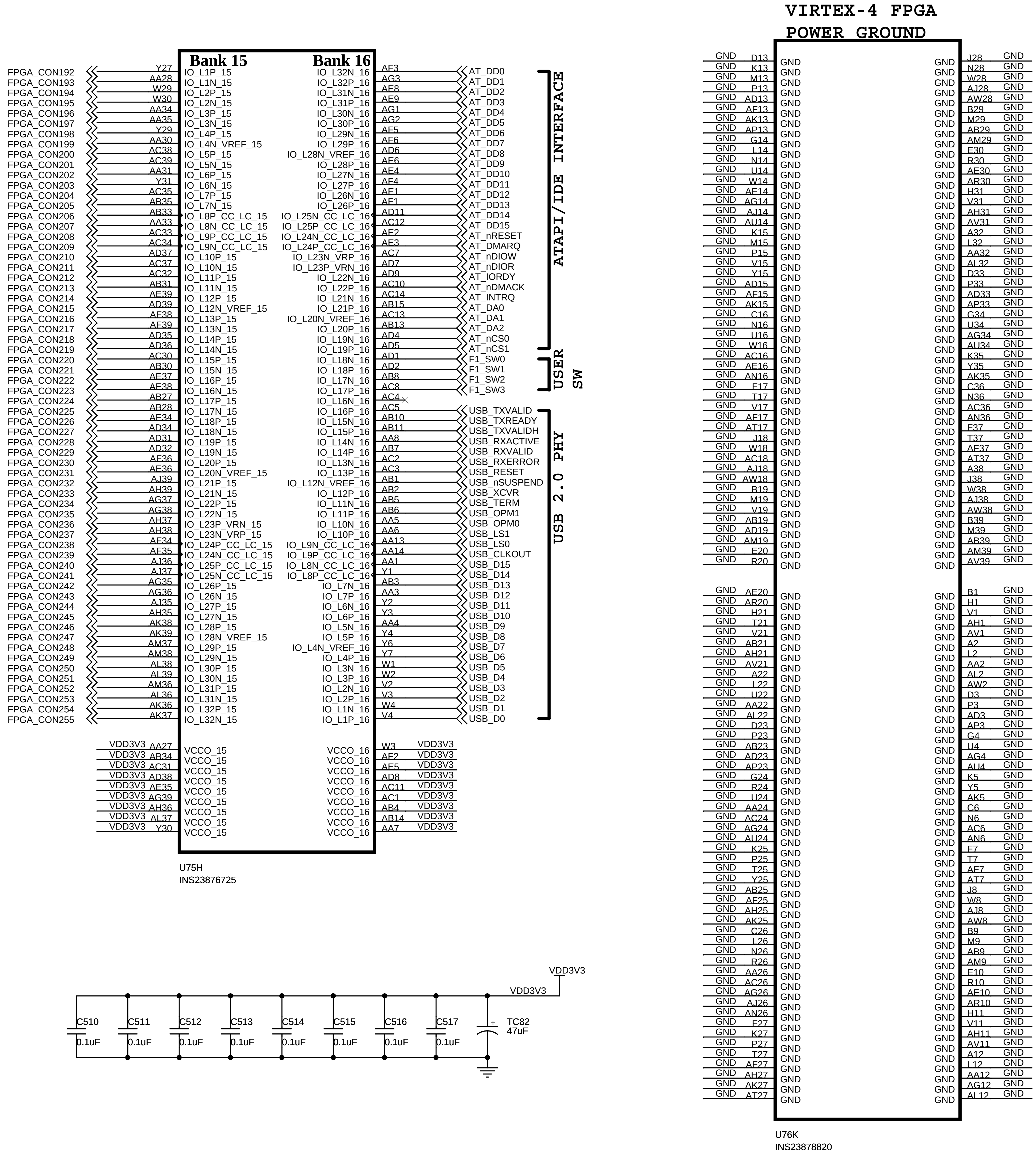


Title	SANGHWA FPGA PLATFORM FPGA1 BANK 7/8/9/10	
Size A3	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 23 of 26	

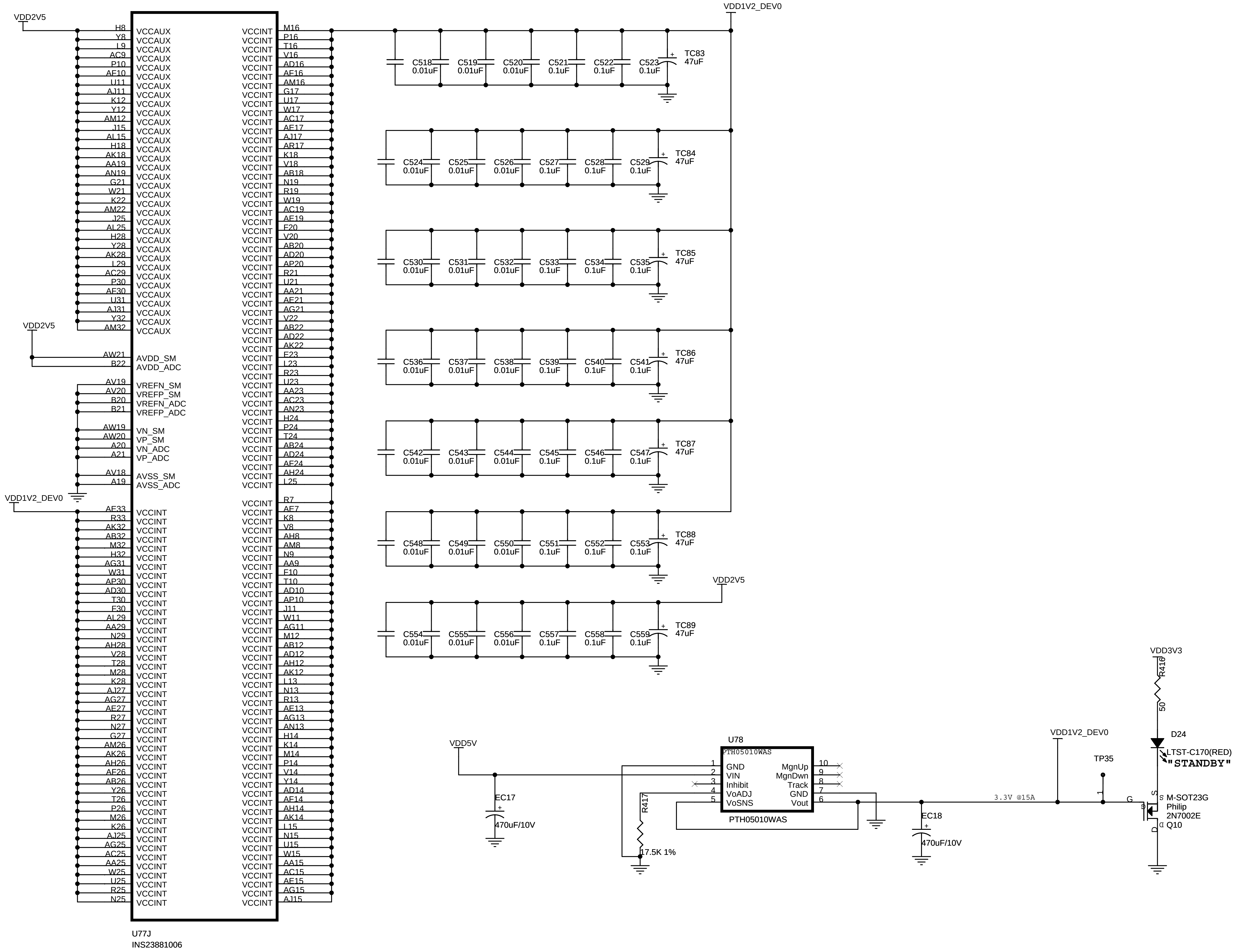


SANGHWA FPGA PLATFORM

Title	FPGA1 BANK 11/12/13/14	
Size A3	Document Number <Doc>	Rev 1.0
Date: Saturday, July 22, 2006	Sheet 24 of 26	



Title	SANGHWA FPGA PLATFORM		
	FPGA1 BANK 15/16/GND		
Size A3	Document Number <Doc>		Rev 1.0
Date: Saturday, July 22, 2006		Sheet 25 of 26	



	SANGHWA FPGA PLATFORM
Title	FPGA1 POWER VCCO
Size A3	Document Number <Doc> — Rev 1.0
Date:	Saturday, July 22, 2006 — Sheet 26 of 26